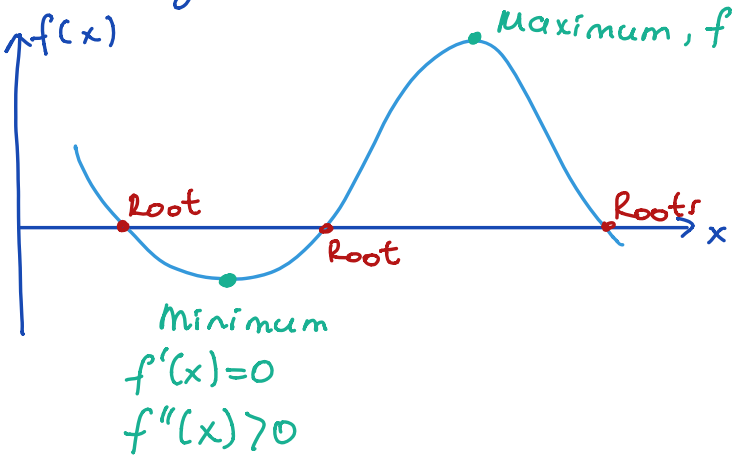
CHAPTER 7 – Optimization

- Mathematical perspective: optimization deals with finding the minima and maxima of a function that depends on one or more variables. The goal is to determine the values of the variables that give the minima or maxima of the function.



Relationship between roots and optima:

- Differentiate the function and locate the root (the zero)

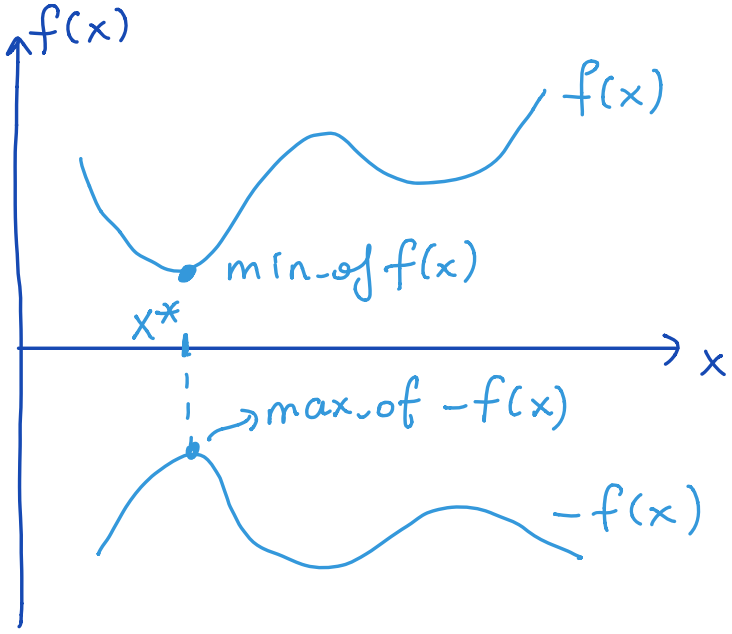
$f'(x)=0$ ($f'(x)$ is generally not available analytically)

- Direct optimization methods are available.

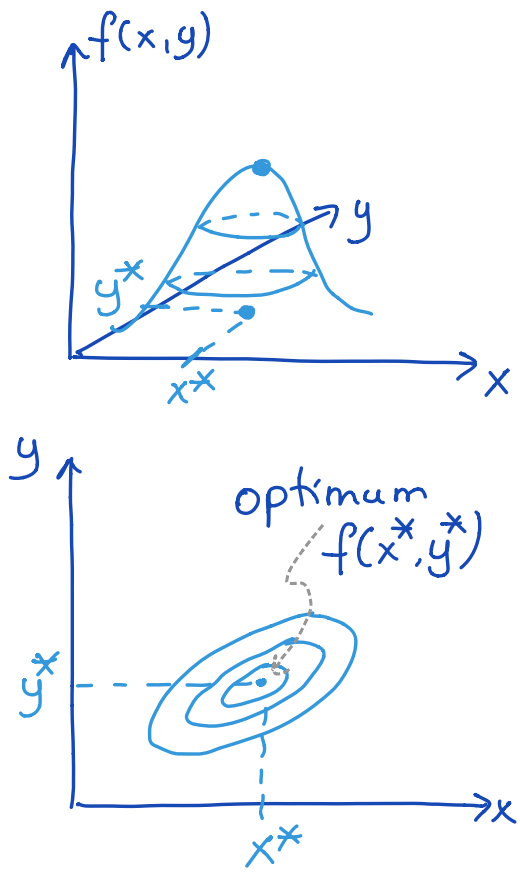
One-dimensional problems: involve functions that depend on a single dependent variable.

Multi-dimensional problems: involve functions that depend on two or more dependent variables.

One-dimensional optimization



2D Optimization



* Finding a maximum or minimum are identical. x^* minimizes $f(x)$ and maximizes $-f(x)$

Optimization – Mathematical Background

An optimization or mathematical programming problem can be stated as:

Find $\underline{x}$, which minimizes or maximizes $f(\underline{x})$ subject to:

$$d_i(\underline{x}) \leq a_i \quad , i = 1, 2, \ldots, m \qquad \underline{x} \in \mathbb{R}^n$$

$$e_i(\underline{x}) = b_i \quad , i = 1, 2, \ldots, p$$

$\underline{x}$: n dimensional vector

$f(\underline{x})$: objective function
$d_i(\underline{x})$: inequality constraints
$e_i(\underline{x})$: equality constraints
a_i, b_i : constants

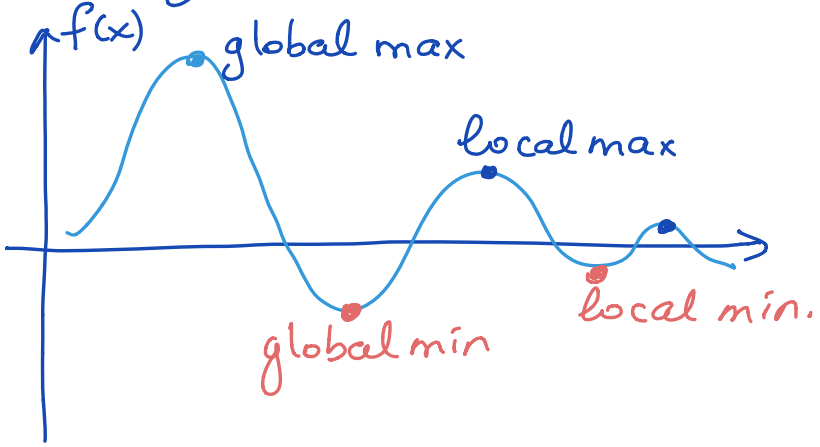
- If $f(\underline{x})$ and the constraints are linear, we have linear programming.
- If $f(\underline{x})$ is quadratic and the constraints are linear, we have quadratic programming.
- If $f(\underline{x})$ is not linear or quadratic and/or the constraints are nonlinear, we have nonlinear programming.
- If constraints are included, we have constrained optimization, otherwise it is an unconstrained optimization problem.
- Constrained problems: Degree of freedom = $n-p-m$

To obtain a solution $p+m \leq n$. If $p+m > n$, the problem is said to be overdetermined.

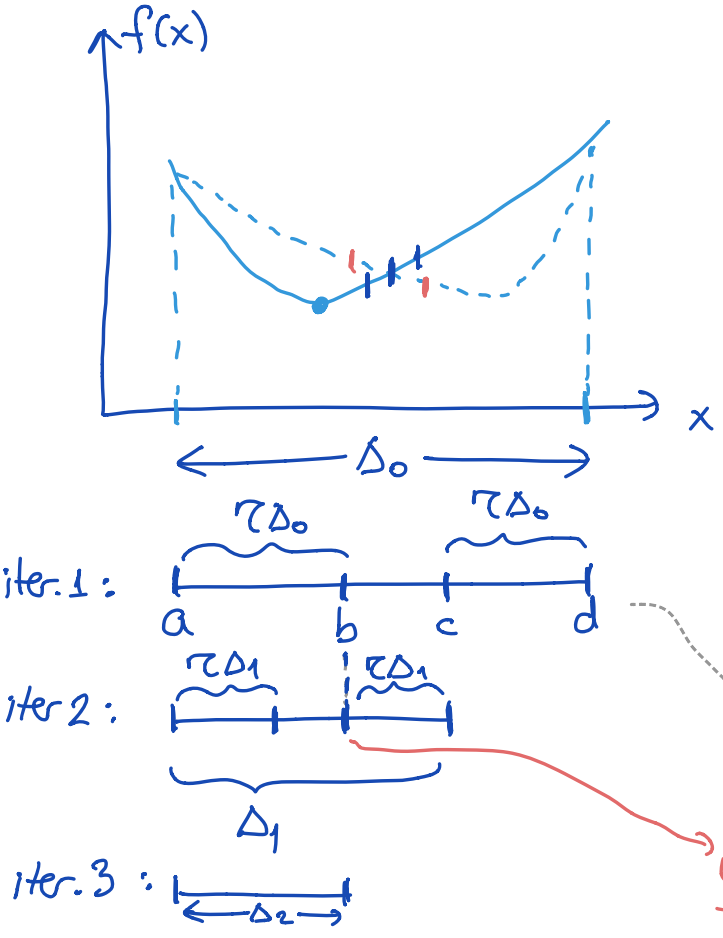
One-dimensional Optimization

Global optimum: the very best solution.
Local optimum: though not the very best, better than its immediate neighbors.

- we are interested in finding the global minimum.
- We do not want to mistake a local optimum for the global optimum.



Golden Section Method (Michael Zibulewsky)



- Evaluating a function at a single intermediate value will not tell whether the min. is in the left half or right half.
- Evaluate at two intermediate pts and get an idea of slope.

if $f(b) < f(c)$
choose ac
else
choose bd

We want to carry pt. b to the next iteration.

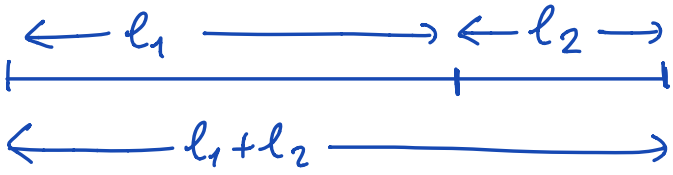
$$\Delta_1 = \Delta_0 - \tau\Delta_0 = (1-\tau)\Delta_0$$

$$\tau\Delta_0 = \Delta_1 - \tau\Delta_1 = (1-\tau)\Delta_1 = (1-\tau)(1-\tau)\Delta_0$$
$$= (1-\tau)^2\Delta_0$$

$$\tau\cancel{\Delta_0} = (1-\tau)^2\cancel{\Delta_0} \quad \Rightarrow \tau = (1-\tau)^2 \Rightarrow \tau = \frac{3-\sqrt{5}}{2} = 0.3820$$

Most efficient method to find the minimum, when you only have function evaluations (not derivative evaluations)

Golden-ratio:



Euclid's golden ratio: divide a line so that

$$\frac{l_1+l_2}{l_1} = \frac{l_1}{l_2}$$

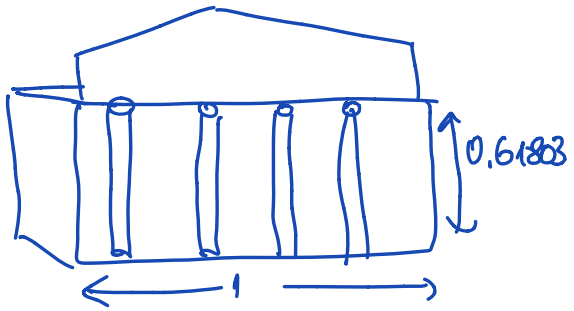
Multiply by $\frac{l_1}{l_2}$ and collect terms:

$$\underbrace{\frac{l_1}{l_2}}_{\phi} + 1 = \frac{l_1^2}{l_2^2}$$

$$\phi = \frac{l_1}{l_2}$$

$$\phi^2 - \phi - 1 = 0 \;, \Rightarrow \phi = \frac{1+\sqrt{5}}{2} = 1.618$$

Parthenon in Athens:



$\phi - 1$: Golden ratio = 0.618

(Golden ratio has been considered as aesthetically pleasing in Western cultures.)

Golden Section Search

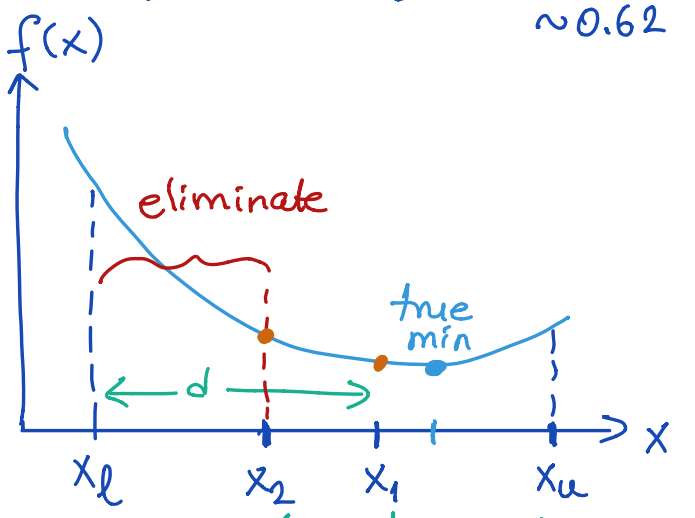
* Start by defining an interval that contains a single answer.

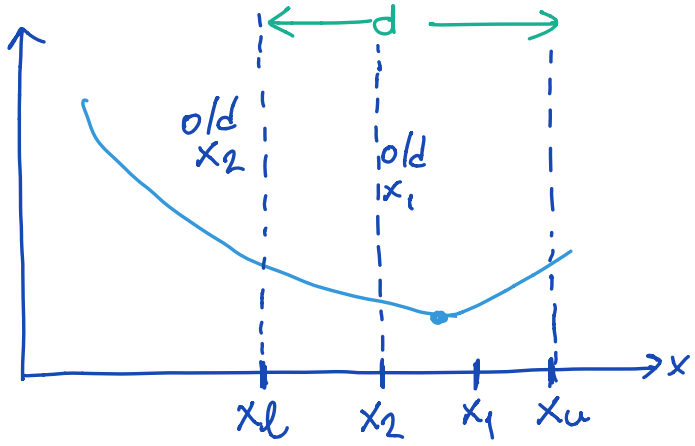
x_l : lower bound

x_u : upper bound

Initial step : Determine intermediate points x_1 and x_2.

$$\left.\begin{aligned} x_1 &= x_l + d \\ x_2 &= x_u - d \end{aligned}\right\} \quad d = \underbrace{(\phi - 1)}_{\sim 0.62}(x_u - x_l)$$





inital step : choose two interior pts. according to the golden ratio.

Second step : Define a new interval that encompasses the optimum

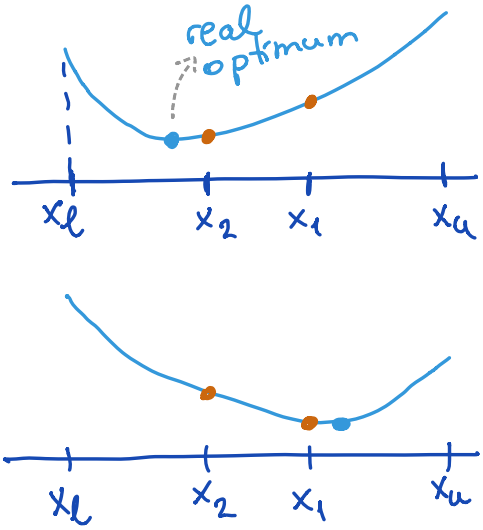
Second Step: Evaluate the function at the two intermediate points.

1) If $f(x_1) < f(x_2)$ then $f(x_1)$ is the minimum
   - eliminate the region from x_l to x_2 since it does not contain the minimum (left of x_2)
   - replace x_l with x_2
   - old x_1 becomes x_2
   - $x_1 = x_l + d_{new}$

2) If $f(x_2) < f(x_1)$, then $f(x_2)$ is the minimum.
   - Eliminate the right of x_1 to x_u.
   - Replace x_u with x_1
   - old x_2 becomes x_1
   - $x_2 = x_u - d_{new}$

Upper Bound for the Error for Golden-Section Search

Once an iteration is complete, the optimum can be in one of the two intervals:



Case 1: current optimum x_2
- real optimum in the lower interval (x_l, x_2, x_1)

Case 2: current optimum at x_1

real optimum in upper interval (x_2, x_1, x_u)

Both cases are similar, look at the upper interval (case 2)

(x_2, x_1, x_u)

$\Rightarrow$ If the true value were at x_2 (far left), max. distance would be: (x_1 = current best)

$$\Delta x_a = x_1 - x_2$$
$$= \underbrace{x_l + (\phi-1)(x_u - x_l)}_{x_1} - \underbrace{[x_u - (\phi-1)(x_u - x_l)]}_{x_2}$$
$$= (x_l - x_u) + 2(\phi-1)(x_u - x_l)$$
$$= (2\phi - 3)(x_u - x_l) = 0.2361\,(x_u - x_l)$$

$\Rightarrow$ If true value were at far right (x_u)

$$\Delta x_b = x_u - x_1 \quad (x_1 = \text{current best})$$
$$= x_u - [x_l + (\phi-1)(x_u - x_l)]$$
$$= (x_u - x_l) - (\phi-1)(x_u - x_l) = (2-\phi)(x_u - x_l)$$
$$= 0.3820\,(x_u - x_l)$$

Normalize to get relative error for that iteration:

$$\varepsilon_a = (2-\phi)\left|\frac{x_u - x_l}{x_{opt}}\right| \times 100\,\%$$

(x_{opt} → optimal value for that iteration)

can be used as a terminating condition.

- At each iteration, the interval is reduced by a factor of about 61.8%.
- After 20 iterations interval shrunk to about 0.0066% of its initial length.

EX Find the minimum of $f(x) = \frac{x^2}{10} - 2\sin x$, from $x_l = 0$ to $x_u = 4$

Sol:

iteration 1

$d = 0.618(x_u - x_l) = 0.618(4-0) = 2.4721$

$x_1 = x_l + d = 0 + 2.4721 = 2.4721$, $f(x_1) = -0.63$

$x_2 = x_u - d = 1.5279$, $f(x_2) = -1.7647$

Since $f(x_2) < f(x_1) \Rightarrow$ best estimate of min. is at $x_2 = 1.5279$

$\Rightarrow$ min. is in the interval x_l, x_2 and x_1

$\rightarrow x_l = 0$, $x_u = x_1 = 2.4721$

$\rightarrow x_1 = x_2 = 1.5279$, $f(x_1) = -1.7647$

$\rightarrow$ update $d = 0.62(2.4721 - 0) = 1.5279$

$\rightarrow x_2 = x_u - d = 2.4721 - 1.5279 = 0.9443$,

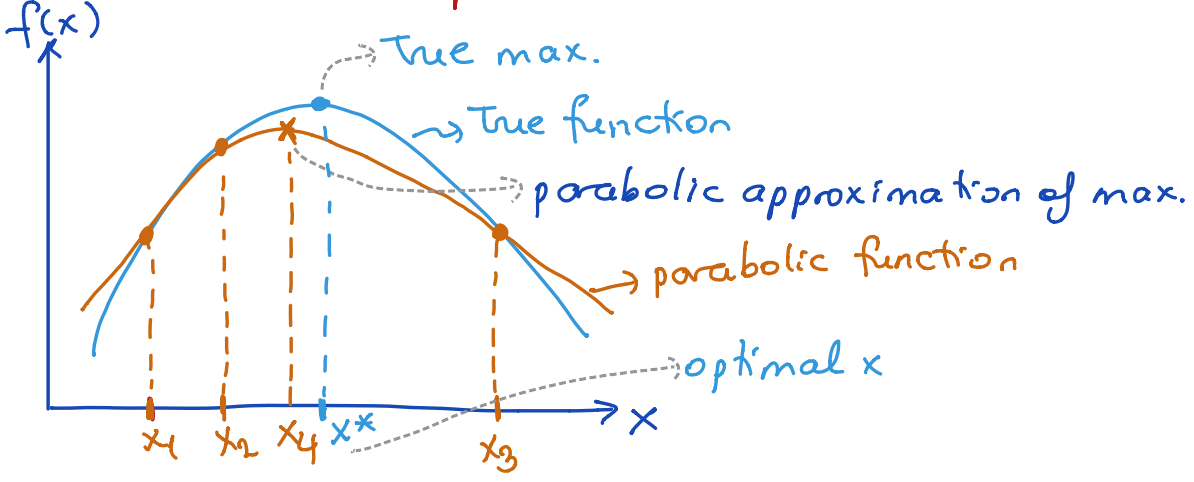
$f(x_2) = -1.5310$

iteration 2

- $f(x_2) > f(x_1) \Rightarrow$ best min. at $x_1 = 1.5279$ in interval (x_2, x_1, x_u)

⋮

Parabolic Interpolation



⇒ Fit a parabola (unique) to the points $(x_1, f(x_1)), (x_2, f(x_2))$ $(x_3, f(x_3))$ ⟶ initial guesses.

⇒ Differentiate it and set the result equal to zero, and solve for an estimate of the optimal x.

$$x_4 = x_2 - \frac{1}{2}\frac{(x_2-x_1)^2[f(x_2)-f(x_3)] - (x_2-x_3)^2[f(x_2)-f(x_1)]}{(x_2-x_1)[f(x_2)-f(x_3)] - (x_2-x_3)[f(x_2)-f(x_1)]} \quad (1)$$

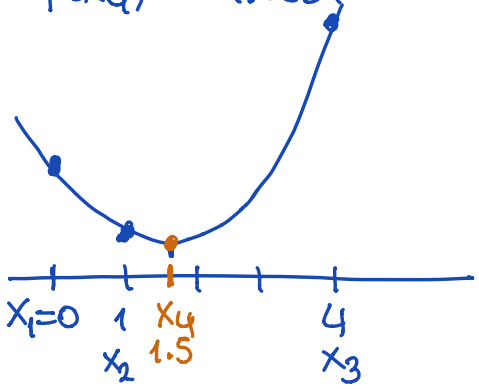
↳ optimum value of the parabolic fit to the guesses (3 initial pts.)

Ex Approximate the min. of $f(x) = \frac{x^2}{10} - 2\sin x$ with initial guesses $x_1 = 0, x_2 = 1, x_3 = 4$

$f(x_1) = 0$, $f(x_2) = -1.58$, $f(x_3) = 3.11$

⇒ $x_4 = 1.50$ from (1)

$f(x_4) = -1.769$



Since $f(x_4) < f(x_2)$, discard x_1

- Set $x_1 = 1$, $x_2 = 1.5$, $x_3 = 4$ in the next iteration.

MATLAB Function : fminbnd

Combines:

- slow, dependable golden section search
- faster, (maybe unreliable) parabolic interpolation.

Syntax:

$[x_{min}, fval] = fminbnd(function, x1, x2)$

x_{min}: location of min

fval: value of min.

x1, x2: bounds of the interval being searched

Newton's Method

- An open method to find the root x of a function such that $f(x)=0$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

- To find the optimum of $f(x)$ define $g(x)=f'(x)$

$$f'(x^*) = g(x^*) = 0$$

$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$$

- we can also use finite-difference approximation for the derivatives.